Palak Harinkhede

Portfolio — palakharinkhede1@gmail.com — 9673494515 — LinkedIn — GitHub — Open to Relocation

Summary

Software Engineer with 1+ year of experience, specializing in full-stack development, AI/ML, and data engineering. Experienced in building scalable applications, REST APIs, agentic AI workflows, predictive models, forecasting pipelines, and anomaly detection systems. Proficient in Python, SQL, machine learning, databases, and end-to-end software development, with a focus on building reliable, production-ready solutions.

Experience

STMicroelectronics, Noida **Jul 2025 – Present**

AI Engineer

- Built multi-agent workflows to automate PLM (Product Lifecycle Management) tasks by routing requests to workflow-specific tools and retrieving relevant enterprise knowledge; integrated Python, LLM function calling, and REST APIs.
- Built classification, regression, and anomaly-detection models for high-dimensional industrial data with Random Forest, XGBoost, and CatBoost; handled preprocessing, feature engineering, model comparison, and tuning.

Personal Projects

Snippet Vault | Full-Stack Snippet Management SaaS GitHub

Tech: Next.js, React, TypeScript, Prisma, PostgreSQL, NextAuth.js, Stripe

- Built a full-stack SaaS for snippet CRUD, tagging, search, and sharing; implemented JWT authentication, server-side plan enforcement, Stripe Checkout, webhooks, and billing portal.

Flockit | Collaborative Habit Tracker Play Store

Tech: React 19, PWA, Capacitor, Android, REST APIs, Neon PostgreSQL, JSONB

- Built an offline-first web, iOS PWA, and Android tracker with reminders, streaks, and collaborative modes; implemented serverless REST APIs and compact PostgreSQL JSONB storage.

Voice Finance Buddy | Multi-Agent Financial AI Assistant GitHub

Tech: Python, GPT-4, Whisper, RAG, FAISS, SQLite, Streamlit

- Built a multilingual voice assistant with persistent memory, guardrails, and function-calling Budget, Expense, Investment, and Tax agents; delivered sub-3-second voice response latency.

DocQuery AI | Enterprise Document Intelligence GitHub

Tech: Python, FAISS, Sentence Transformers, TinyLlama, FastAPI, Streamlit

- Built a local document RAG system with semantic chunking, query expansion, vector retrieval, and grounded generation without external APIs.

MoodBeat AI | Emotion-Aware Music Recommendation GitHub

Tech: Python, MediaPipe, PyTorch, EfficientNet-B5, FastAPI, OpenCV

- Built a real-time facial emotion and music recommendation pipeline using 468 MediaPipe landmarks, smoothed tracking, and a fine-tuned four-class EfficientNet-B5 model.

Technical Skills

Programming & Data: Python, C++, TypeScript, SQL, GoogleSQL, Pandas, NumPy, BigQuery

Software Engineering: Data Structures & Algorithms, OOP, modular design, debugging, REST APIs, Next.js, React, FastAPI, Streamlit, Tailwind CSS, Prisma, JWT authentication, Stripe, Docker, CI/CD, Git, Linux, Vercel, Capacitor, PWA

Databases: PostgreSQL, SQLite, JSONB

Data Science & Analytics: Statistical modeling, classification, regression, feature engineering, model evaluation, hyperparameter tuning, time-series forecasting, anomaly detection, Prophet, Scikit-learn, XGBoost, CatBoost, Excel

AI & Retrieval: LLMs, Agentic AI, multi-agent systems, RAG, function calling, MCP, NLP, PyTorch, TensorFlow, computer vision, OpenCV, Vector Databases (FAISS), Sentence Transformers, Hugging Face Transformers, semantic/vector search, query expansion

Education

Delhi Technological University **2024 – 2026**

M.Tech in Computer Science Engineering CGPA: 8.5/10

Sinhgad College of Engineering, Pune **2018 – 2022**

B.E. in Computer Science Engineering; Honours in AI & ML CGPA: 8.85/10

Certifications

Microsoft Azure AI Essentials; Career Essentials in Generative AI (Microsoft & LinkedIn); Career Essentials in GitHub; EF SET English 88/100 (C2 Proficient)